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# AQA A Level Further Mathematics (7367)

Topic **DB: Networks**

Sub topic **DB5: Evaluate, Modify, and Refine Models**

## Topic Questions

# AQA A Level Further Mathematics (7367)

## Question Paper

Topic DB: Networks

Sub topic DB5: Evaluate, Modify, and Refine Models



**To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.**

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**Name :** ____________________

**Date :** ____________________

**Time :** ____________________

**Total Marks Available :** _____

**Total Marks Received :** _____

## Q1.

A communications company is conducting a feasibility study into the installation of underground television cables between 5 neighbouring districts.

The length of the possible pathways for the television cables between each pair of districts, in miles, is shown in the table.

The pathways all run alongside cycle tracks.

| | **Billinge** | **Garswood** | **Haydock** | **Orrell** | **Up Holland** |
|---|---|---|---|---|---|
| **Billinge** | - | 2.5 | *** | 4.3 | 4.8 |
| **Garswood** | 2.5 | - | 3.1 | *** | 5.9 |
| **Haydock** | *** | 3.1 | - | 6.7 | 7.8 |
| **Orrell** | 4.3 | *** | 6.7 | - | 2.1 |
| **Up Holland** | 4.8 | 5.9 | 7.8 | 2.1 | - |

(a) Give a possible reason, in context, why some of the table entries are labelled as ***.

**(1)**

(b) As part of the feasibility study, Sally, an engineer, needs to assess each possible pathway between the districts. To do this, Sally decides to travel along every pathway using a bicycle, starting and finishing in the same district. From past experience, Sally knows that she can travel at an average speed of 12 miles per hour on a bicycle.



Find the minimum time, in minutes, that it will take Sally to cycle along every pathway.

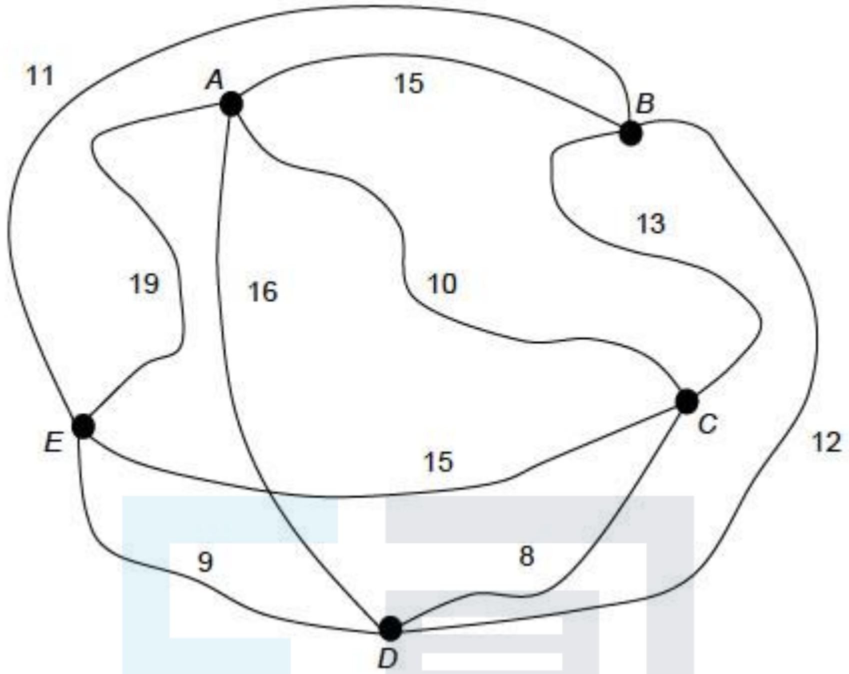
**(4)**

**(Total 5 marks)**

## Q2.

Charlotte is visiting a city and plans to visit its five monuments: *A, B, C, D* and *E*.

The network shows the time, in minutes, that a typical tourist would take to walk between the monuments on a busy weekday morning.




Charlotte intends to walk from one monument to another until she has visited them all, before returning to her starting place.

(a) Use the nearest neighbour algorithm, starting from $A$, to find an upper bound for the minimum time for Charlotte's tour.

**(2)**

(b) By deleting vertex $B$, find a lower bound for the minimum time for Charlotte's tour.

**(3)**



(c) Charlotte wants to complete the tour in 52 minutes. Use your answers to parts **(a)** and **(b)** to comment on whether this could be possible.

**(1)**

(d) Charlotte takes 58 minutes to complete the tour. Evaluate your answers to part **(a)** and part **(b)** given this information.

**(1)**

(e) Explain how this model for a typical tourist's tour may not be applicable if the tourist walked between the monuments during the evening.

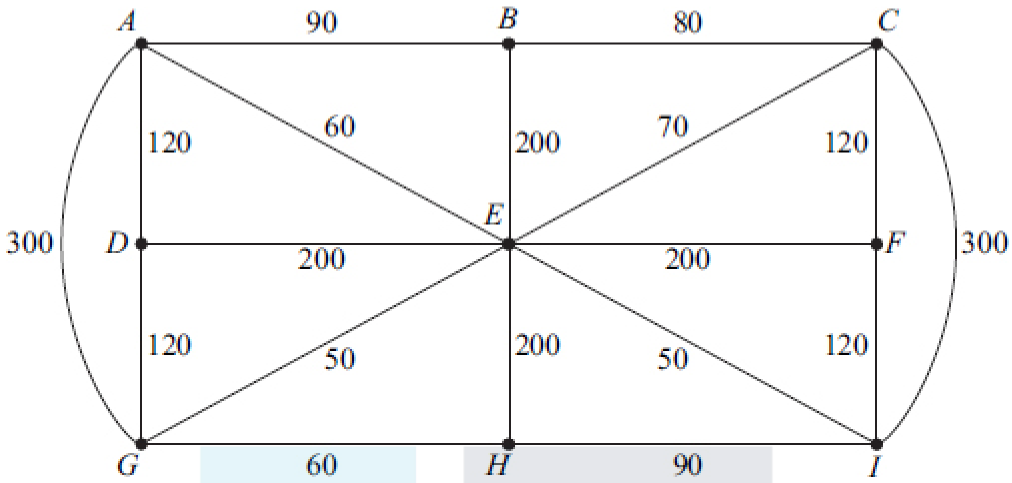
**(1)**

**(Total 8 marks)**

## Q3.

The network below shows some streets in a town. The number on each edge shows the length of that street, in metres.

Leaflets are to be distributed by a restaurant owner, Tony, from his restaurant located at vertex $B$. Tony must start from his restaurant, walk along all the streets at least once, before returning to his restaurant.




The total length of the streets is 2430 metres.

(a) Find the length of an optimal Chinese postman route for Tony.

**(5)**

(b) Colin also wishes to distribute some leaflets. He starts from his house at $H$, walks along all the streets at least once, before finishing at the restaurant at $B$.

Colin wishes to walk the minimum distance. Find the length of an optimal route for Colin.

**(1)**

(c) David also walks along all the streets at least once. He can start at any vertex and finish at any vertex. David also wishes to walk the minimum distance.



(i) Find the length of an optimal route for David.

**(1)**

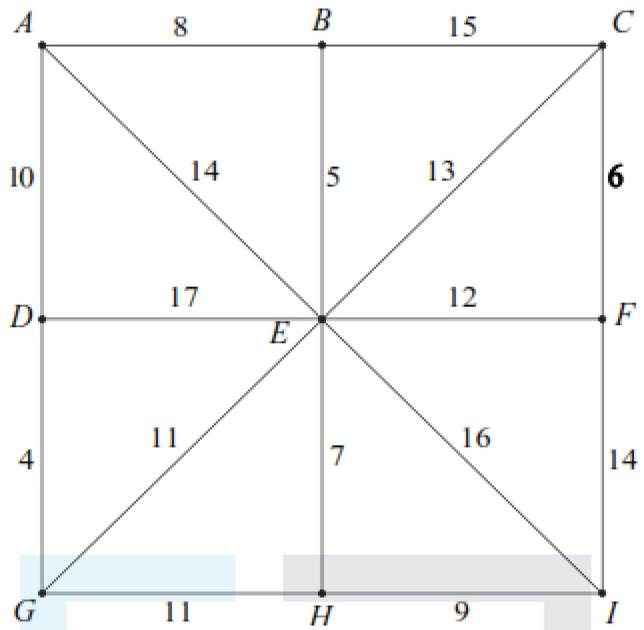
(ii) State the vertices from which David could start in order to achieve this optimal route.

**(1)**

**(Total 8 marks)**

## Q4.

The following network shows the lengths, in miles, of roads connecting nine villages, $A, B, \ldots, I$.




(a) (i) Use Prim's algorithm starting from $E$, showing the order in which you select the edges, to find a minimum spanning tree for the network.

**(4)**

(ii) State the length of your minimum spanning tree.

**(1)**

(iii) Draw your minimum spanning tree.

**(2)**

(b) On a particular day, village $B$ is cut off, so its connecting roads cannot be used.

Find the length of a minimum spanning tree for the remaining eight villages.



**(2)**

**(Total 9 marks)**

## Q5.

A group of eight friends, $A$, $B$, $C$, $D$, $E$, $F$, $G$ and $H$, keep in touch by sending text messages. The cost, in pence, of sending a message between each pair of friends is shown in the following table.

| | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ | $H$ |
|---|---|---|---|---|---|---|---|---|
| $A$ | – | 15 | 10 | 12 | 16 | 11 | 14 | 17 |
| $B$ | 15 | – | 15 | 14 | 15 | 16 | 16 | 15 |
| $C$ | 10 | 15 | – | 11 | 10 | 12 | 14 | 9 |
| $D$ | 12 | 14 | 11 | – | 11 | 12 | 14 | 12 |
| $E$ | 16 | 15 | 10 | 11 | – | 13 | 15 | 14 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ***F*** | 11 | 16 | 12 | 12 | 13 | – | 14 | 8 |
| ***G*** | 14 | 16 | 14 | 14 | 15 | 14 | – | 13 |
| ***H*** | 17 | 15 | 9 | 12 | 14 | 8 | 13 | – |

One of the group wishes to pass on a piece of news to all the other friends, either by a direct text or by the message being passed on from friend to friend, at the minimum total cost.

(a) (i) Use Prim's algorithm starting from $A$, showing the order in which you select the edges, to find a minimum spanning tree for the table.

**(4)**

(ii) Draw your minimum spanning tree.

**(2)**

(iii) Find the minimum total cost.

**(1)**

(b) Person $H$ leaves the group. Find the new minimum total cost.

**(2)**

**(Total 9 marks)**

## Q6.

Angelo is visiting six famous places in Palermo: $A$, $B$, $C$, $D$, $E$ and $F$. He intends to travel from one place to the next until he has visited all of the places before returning to his starting place. Due to the traffic system, the time taken to travel between two places may be different dependent on the direction travelled.



The table shows the times, in minutes, taken to travel between the six places.

| From \ To | ***A*** | ***B*** | ***C*** | ***D*** | ***E*** | ***F*** |
|---|---|---|---|---|---|---|
| ***A*** | – | 25 | 20 | 20 | 27 | 25 |
| ***B*** | 15 | – | 10 | 11 | 15 | 30 |
| ***C*** | 5 | 30 | – | 15 | 20 | 19 |
| ***D*** | 20 | 25 | 15 | – | 25 | 10 |
| ***E*** | 10 | 20 | 7 | 15 | – | 15 |
| ***F*** | 25 | 35 | 29 | 20 | 30 | – |

(a) Give an example of a Hamiltonian cycle in this context.

**(2)**

(b) (i) Show that, if the nearest neighbour algorithm starting from $F$ is used, the total travelling time for Angelo would be 95 minutes.

**(3)**

(ii) Explain why your answer to part (b)(i) is an upper bound for the minimum travelling time for Angelo.

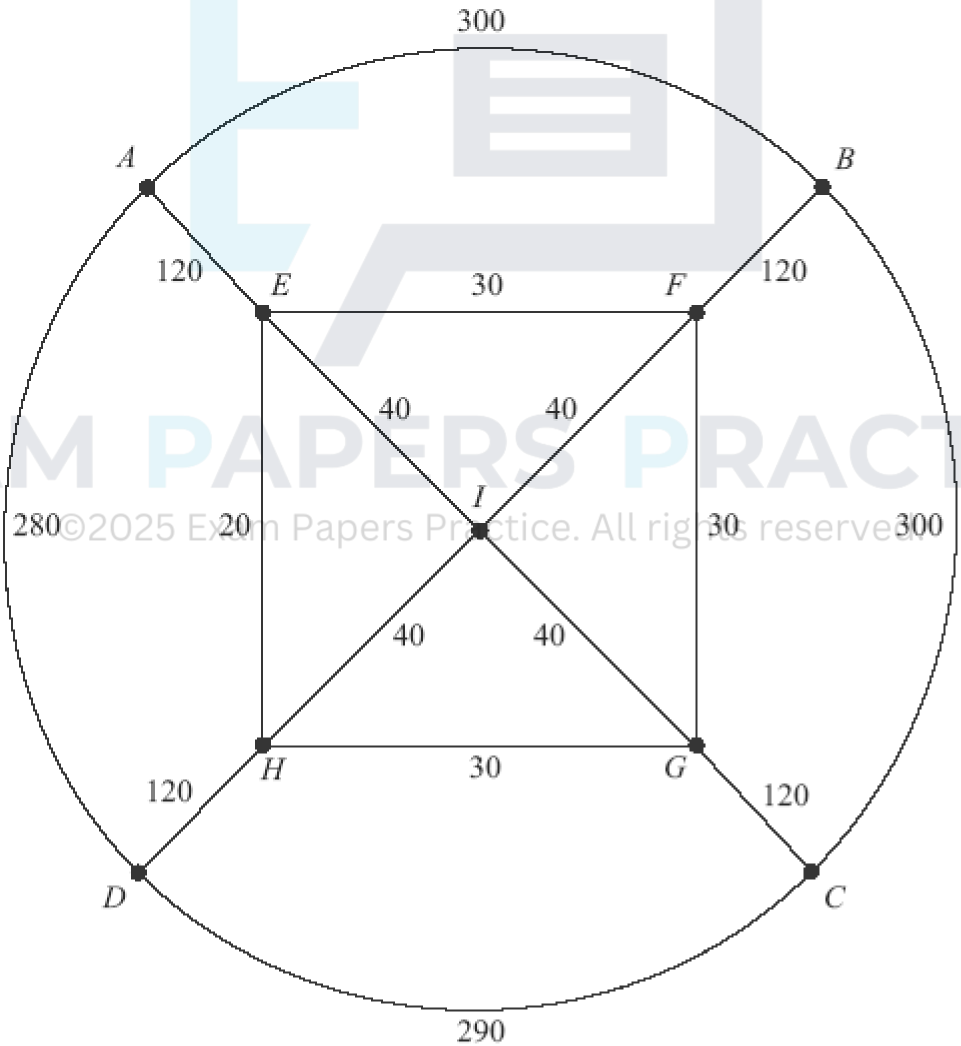
**(2)**

(c) Angelo starts from $F$ and visits $E$ next. He also visits $B$ before he visits $D$. Find an improved upper bound for Angelo's total travelling time.

**(3)**
**(Total 10 marks)**

## Q7.

The diagram shows a network of sixteen roads on a housing estate. The number on each edge is the length, in metres, of the road. The total length of the sixteen roads is 1920 metres.






Total Length = 1920 metres

(a) Chris, an ice-cream salesman, travels along each road at least once, starting and finishing at the point $A$. Find the length of an optimal 'Chinese postman' route for Chris.

**(6)**

(b) Pascal, a paperboy, starts at $A$ and walks along each road at least once before finishing at $D$. Find the length of an optimal route for Pascal.

**(2)**

(c) Millie is to walk along all the roads at least once delivering leaflets. She can start her journey at any point and she can finish her journey at any point.

(i) Find the length of an optimal route for Millie.

**(2)**

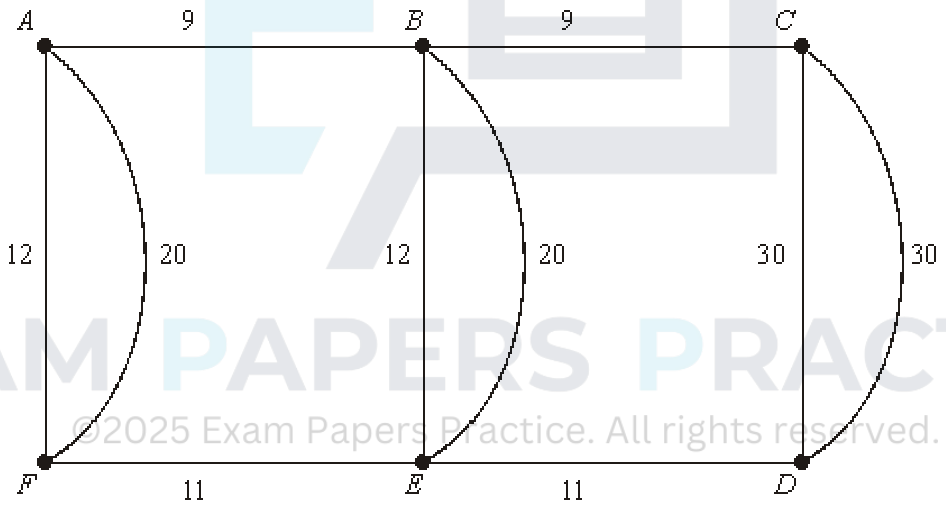
(ii) State the points from which Millie could start in order to achieve this optimal route.

**(1)**

**(Total 11 marks)**

## Q8.

The diagram shows a network of roads connecting 6 villages. The number on each edge is the length, in miles, of the road.






Total length of the roads = 164 miles

(a) A police patrol car based at village $A$ has to travel along each road at least once before returning to $A$. Find the length of an optimal 'Chinese postman' route for the police patrol car.

**(6)**

(b) A council worker starts from $A$ and travels along each road at least once before finishing at $C$. Find the length of an optimal route for the council worker.

**(2)**

(c) A politician is to travel along all the roads at least once. He can start his journey at any village and can finish his journey at any village.

(i) Find the length of an optimal route for the politician.

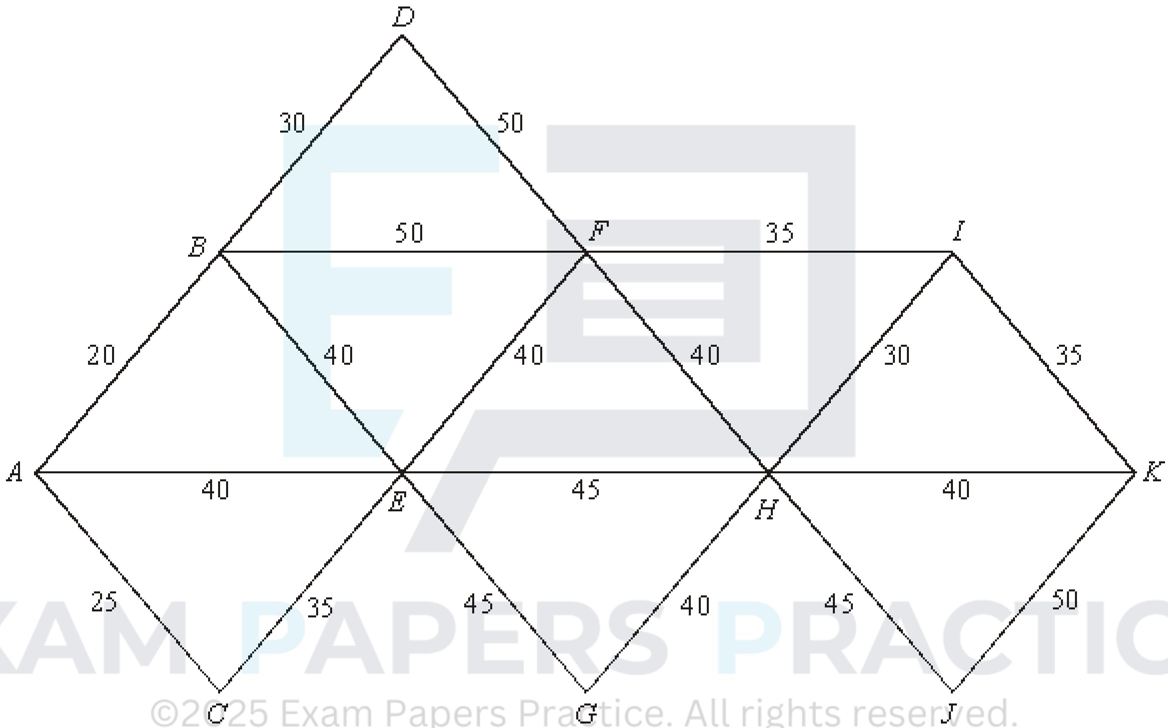
**(2)**

(ii) State the vertices from which the politician could start in order to achieve this optimal route.

**(1)**
**(Total 11 marks)**

## Q9.

A theme park has 11 rides, $A, B, \ldots K$. The network shows the distances, in metres, between pairs of rides. The rides are to be connected by cabling so that information can be collated. The manager of the theme park wishes to use the minimum amount of cabling.






(a) Use Prim's algorithm, starting from $A$, to find the minimum spanning tree for the network.

**(5)**

(b) State the length of cabling required.

**(1)**

(c) Draw your minimum spanning tree.

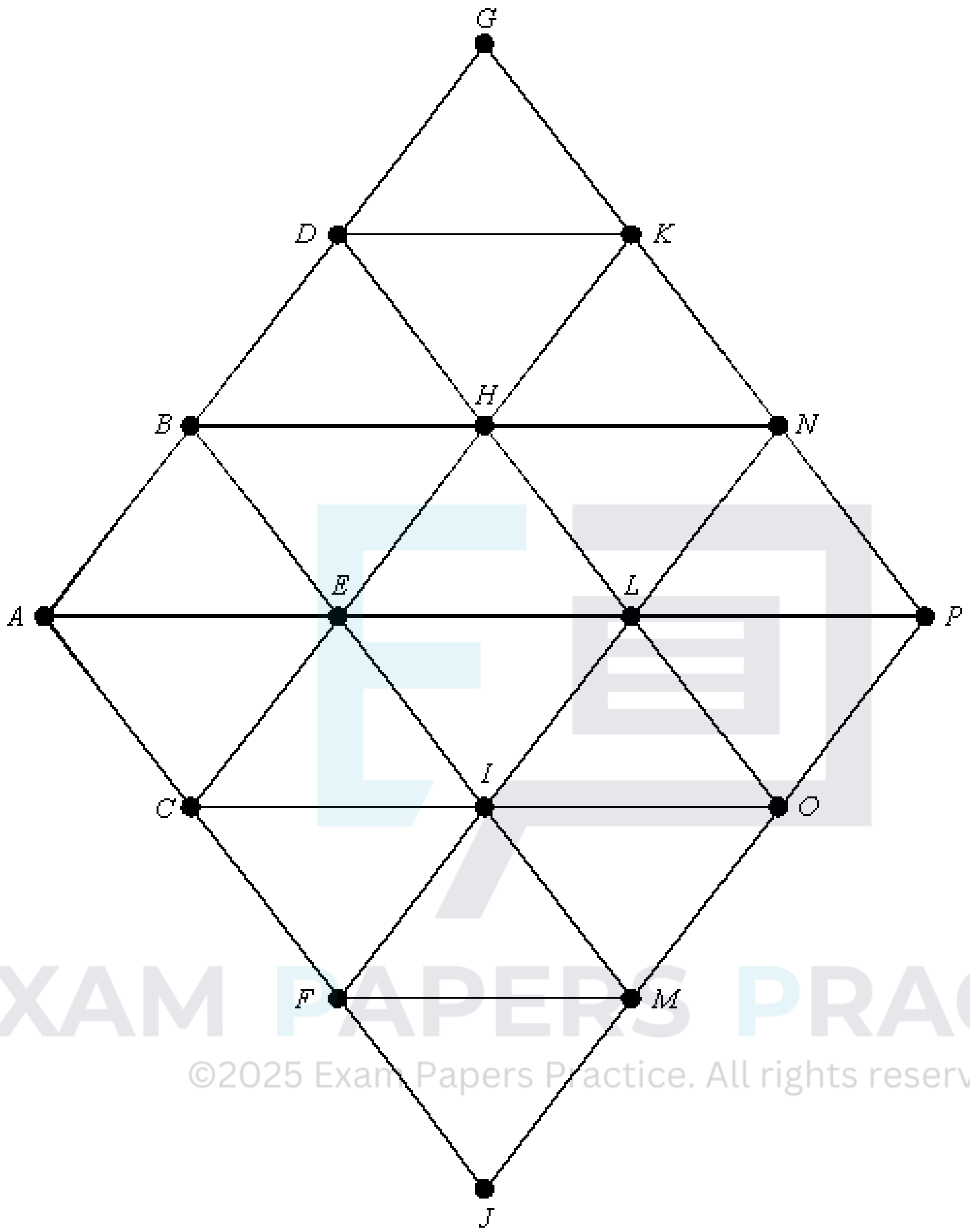
**(3)**

(d) The manager decides that the edge $AE$ must be included. Find the extra length of cabling now required to give the smallest spanning tree that includes $AE$.

**(2)**
**(Total 11 marks)**

## Q10.

The following network has 16 vertices.




(a) Given that the length of each edge is 1 unit, find:

(i) the shortest distance from $A$ to $K$;

**(1)**

(ii) the length of a minimum spanning tree.

**(2)**

(b) (i) Find the length of an optimal Chinese Postman route, starting and finishing at $A$.

**(4)**

(ii) For such a route, state the edges that would have to be used twice.

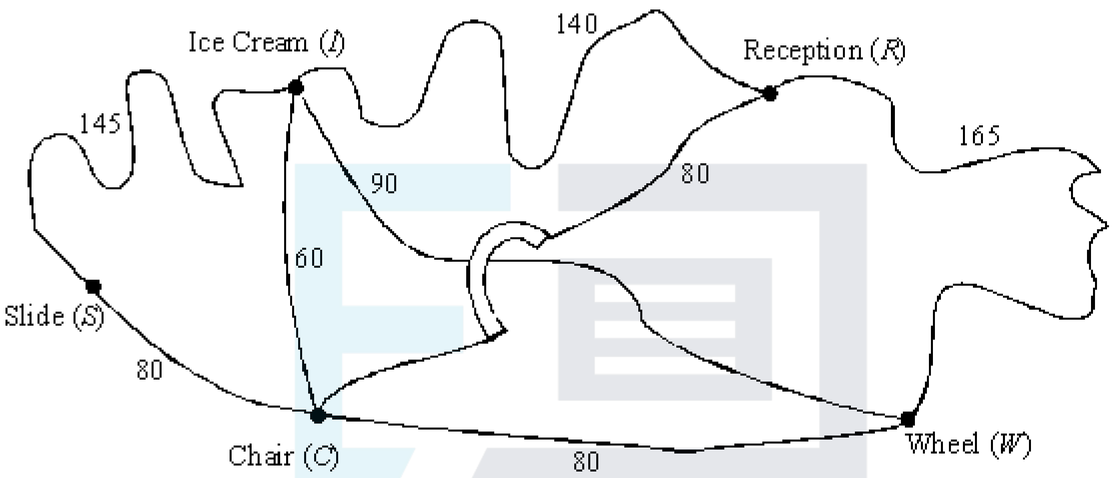
**(1)**

(iii) Given that the edges $AE$ and $LP$ are now removed, find the new length of an optimal Chinese Postman route, starting and finishing at $A$.

**(2)**
**(Total 10 marks)**

## Q11.

A theme park employs a student to patrol the paths and collect litter. The paths that she has to patrol are shown in the following diagram, where all distances are in metres. The path connecting $I$ and $W$ passes under the bridge which carries the path connecting $C$ and $R$.




(a) (i) Find an optimal “Chinese postman” route that the student should take if she is to start and finish at Reception (R).

**(2)**

(ii) State the length of your route.

**(1)**



(b) (i) A service path is to be constructed. Write down the two places that this path should connect, if the student is to be able to walk along every path without having to walk along any path more than once.

**(2)**

(ii) The distance walked by the student in part (b)(i) is shorter than that found in part (a)(ii). Given that the length of the service path is $l$ metres, where $l$ is an integer, find the greatest possible value of $l$.

**(2)**
**(Total 7 marks)**

## Q12.

A group of seven friends, $P$, $Q$, $R$, $S$, $T$, $U$ and $V$, keep in touch by phone. The minimum cost, in pence, of a phone call between any pair of them is shown in the following table.

| | $P$ | $Q$ | $R$ | $S$ | $T$ | $U$ | $V$ |
|---|---|---|---|---|---|---|---|
| $P$ | – | 30 | 50 | 45 | 30 | 35 | 60 |
| $Q$ | 30 | – | 40 | 40 | 80 | 70 | 55 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ***R*** | 50 | 40 | – | 45 | 80 | 70 | 70 |
| ***S*** | 45 | 40 | 45 | – | 35 | 60 | 60 |
| ***T*** | 30 | 80 | 80 | 35 | – | 30 | 60 |
| ***U*** | 35 | 70 | 70 | 60 | 30 | – | 50 |
| ***V*** | 60 | 55 | 70 | 60 | 60 | 50 | – |

Person $P$ wishes to pass on a piece of news to all the other friends, either by a direct phone call or by the message being passed on from friend to friend.

(a) By applying Prim's algorithm to the matrix in the figure above, find the minimum cost of notifying all the friends of the news.

**(5)**

(b) Draw a graph representing the minimum connector which you found in part (a).

**(2)**

(c) Person $Q$ is on holiday and so a message can be left on her answerphone but she cannot be used to pass the message on. In these new circumstances, what is the minimum cost of notifying all the friends of the news?

**(2)**

**(Total 9 marks)**



**Q13.**

The walking times, in minutes, between six points $A$ – $F$ are given in the following table:



| | ***A*** | ***B*** | ***C*** | ***D*** | ***E*** | ***F*** |
|---|---|---|---|---|---|---|
| ***A*** | – | 20 | 30 | 20 | 15 | 10 |
| ***B*** | 20 | – | 35 | 30 | 25 | 25 |
| ***C*** | 30 | 35 | – | 30 | 30 | 25 |
| ***D*** | 20 | 30 | 30 | – | 15 | 25 |
| ***E*** | 15 | 25 | 30 | 15 | – | 10 |
| ***F*** | 10 | 25 | 25 | 25 | 10 | – |

(a) Use Prim's algorithm on the table above, starting at $A$, to find a minimum connector of the six points.

**(5)**

(b) Draw a tree representing the minimum connector which you found in part (a).

**(2)**

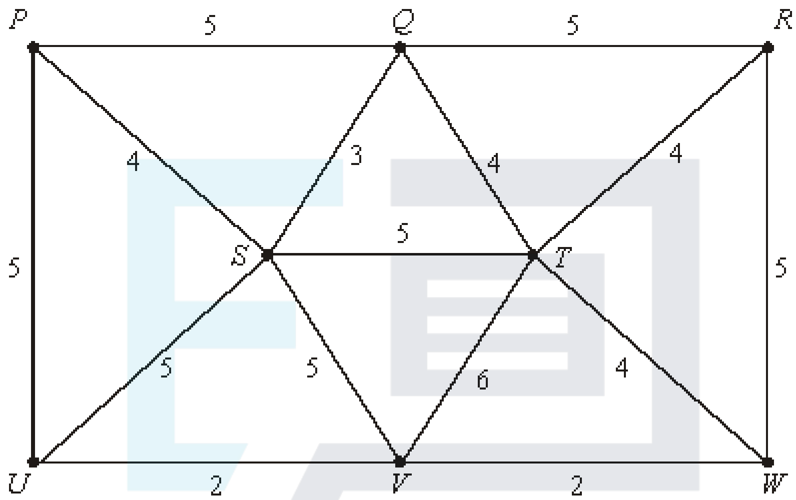
(c) Due to poor weather, only the footpaths in the minimum connector are open. Use your tree from part (b) to show that it is still possible to walk from any one of the six points to any other in less than 1 hour.

**(3)**

**(Total 10 marks)**

## Q14.

The network shows eight Chinese railway stations $P – W$ and the lengths, in miles, of the tracks linking them:




(a) The Chinese Trackrail inspector wishes to travel by rail and check the whole network by starting at $P$, travelling along each track at least once, and ending back at $P$.

(i) Explain why it will be necessary to repeat at least three tracks.



**(2)**

(ii) The inspector wishes to minimise the distance he has to travel. Determine which tracks he must repeat, given that they will include $PS$ or $PU$.

**(5)**

(b) Find the length of the minimum connector of the network and illustrate this minimum connector as a tree.

**(4)**

(c) A train-spotter has to travel exactly 25 miles to get from his home to each of the stations $P$–$W$. He wishes to travel from home to a station, then by train to another station and another and so on until he has visited all the stations once. Then he will return home.

(i) Use your answer to part (b) to show that his round trip must be at least 73 miles in length.

**(2)**

(ii) Explain why a route of 73 miles is not possible and, by adapting your minimum connector, find a round trip of length74 miles for the train-spotter.

**(3)**

**(Total 16 marks)**

## Q15.

The following table shows the lengths, in miles, of pipe-lines between six water-pumping stations $A – F$.

| | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ |
|---|---|---|---|---|---|---|
| $A$ | — | 23 | 17 | 19 | 18 | 22 |
| $B$ | 23 | — | 20 | 24 | 23 | 21 |
| $C$ | 17 | 20 | — | 24 | 19 | 23 |
| $D$ | 19 | 24 | 24 | — | 20 | 24 |
| $E$ | 18 | 23 | 19 | 20 | — | 24 |
| $F$ | 22 | 21 | 23 | 24 | 24 | — |

(a) The authorities decide to disconnect some of the pipe-lines. Use Prim's algorithm on above, starting at $A$, to find the minimum total length of pipe-lines which need to be kept open in order for all the pumping stations to remain linked together.

**(5)**

(b) The pipe-line from $B$ to $C$ is found to be faulty. So when choosing the pipe-lines to keep open the authorities must not include $BC$. By how many miles does this increase the total length of pipe-lines which need to be kept open?

**(3)**

**(Total 8 marks)**

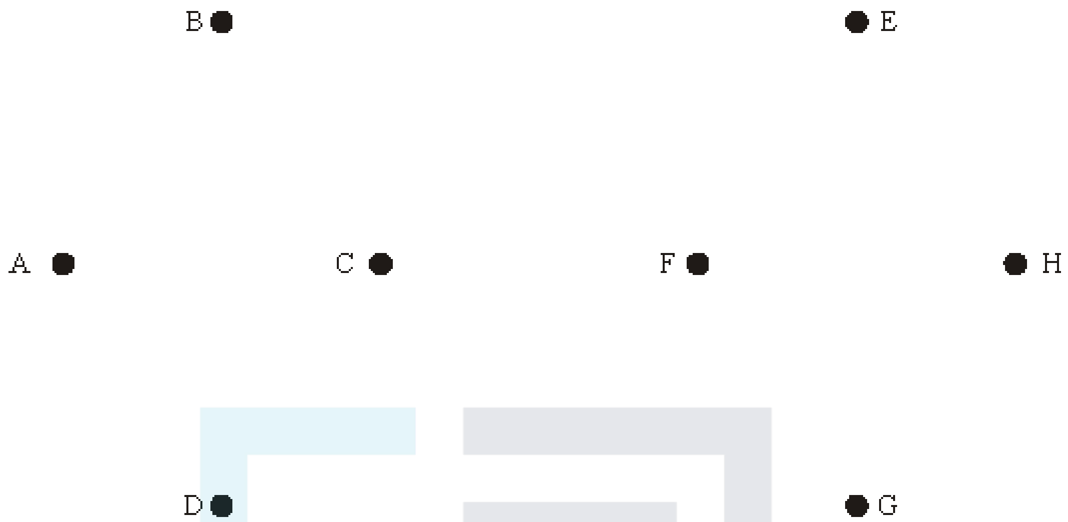


## Q16.

An international organisation has offices $A$, $B$, $C$, $D$, $E$, $F$, $G$ and $H$. The table shows the cost, in pounds, of transmitting a piece of information from one of the offices to another along all existing direct links.

| From \ To | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ | $H$ |
|---|---|---|---|---|---|---|---|---|
| $A$ | – | 12 | 8 | 9 | – | – | – | – |
| $B$ | 12 | – | 6 | – | 10 | – | – | – |
| $C$ | 8 | 6 | – | 10 | – | 10 | – | – |
| $D$ | 9 | – | 10 | – | – | – | 12 | – |
| $E$ | – | 10 | – | – | – | 11 | – | 20 |
| $F$ | – | – | 10 | – | 11 | – | 18 | 18 |
| $G$ | – | – | – | 12 | – | 18 | – | 14 |
| $H$ | – | – | – | – | 20 | 18 | 14 | – |

(a) Construct a network using the vertices $A$ to $H$ below to represent the information in the table.




**(2)**

(b) A piece of information has to be passed from office $A$ to all the other offices, either directly or by being passed on from office to office. Use Kruskal's algorithm to find the minimum total cost of passing the information to all the offices.

**(5)**

(c) Office $C$ joins a communications discount scheme in which the cost of passing information from $C$ is halved (but it does not affect the cost of $C$'s incoming information).

(i) Describe how to adapt the above table to show the revised costs.

**(1)**



(ii) Adapt your answer to part (b) in order to calculate the minimum cost of passing a piece of information from office $A$ to all the other offices with these revised costs.

**(4)**

**(Total 12 marks)**

## Q17.

Some of the eight towns $A$ to $H$ are directly linked by roads. The table shows the distances along these roads in miles.

| | *A* | *B* | *C* | *D* | *E* | *F* | *G* | *H* |
|---|---|---|---|---|---|---|---|---|
| *A* | — | 10 | 9 | 5 | 7 | 6 | 7 | — |
| *B* | 10 | — | 7 | 8 | — | — | 8 | 9 |
| *C* | 9 | 7 | — | — | 7 | 8 | — | 8 |
| *D* | 5 | 8 | — | — | — | 9 | 9 | — |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ***E*** | 7 | — | 7 | — | — | 8 | 9 | 7 |
| ***F*** | 6 | — | 8 | 9 | 8 | — | 6 | 9 |
| ***G*** | 7 | 8 | — | 9 | 9 | 6 | — | — |
| ***H*** | — | 9 | 8 | — | 7 | 9 | — | — |

(a) In the winter the local authority grits some of these roads. They wish to grit the minimum total length of roads possible in order to be able to travel between any two of the towns on gritted roads. Use Prim's algorithm starting at $A$ to find which roads they should grit, and state the minimum total length of roads which need to be gritted.

**(6)**

(b) Illustrate in a graph the minimum connector which you found in (a). How far is it from $A$ to $B$ in your graph?

**(3)**

(c) The ambulance station is at $A$ and next year the local authority decides that it would like to grit the minimum length of roads possible so that the ambulance can reach any of the other towns on gritted roads by travelling less than 15 miles. Show how to adapt your graph in (b) to find the roads which the local authority should grit next year.

**(3)**

**(Total 12 marks)**



## Q18.

A world-wide family consists of people **A, B, C, D, E, F** and **G**. The minimum cost of a phone-call, in pence, between any pair of them is shown in the following table:



| | **A** | **B** | **C** | **D** | **E** | **F** | **G** |
|---|---|---|---|---|---|---|---|
| **A** | – | 80 | 20 | 95 | 10 | 20 | 10 |
| **B** | 80 | – | 70 | 45 | 60 | 70 | 80 |
| **C** | 20 | 70 | – | 80 | 60 | 25 | 25 |
| **D** | 95 | 45 | 80 | – | 70 | 80 | 90 |
| **E** | 10 | 60 | 60 | 70 | – | 25 | 15 |
| **F** | 20 | 70 | 25 | 80 | 25 | – | 30 |
| **G** | 10 | 80 | 25 | 90 | 15 | 30 | – |

Person **A** wishes to pass a piece of news to all the other family members, either by a direct phone-call or by the message being passed on by other phone calls.

(a) By applying Prim's algorithm to the matrix above (or otherwise), find the minimum cost of notifying the whole family of the news.

**(5)**

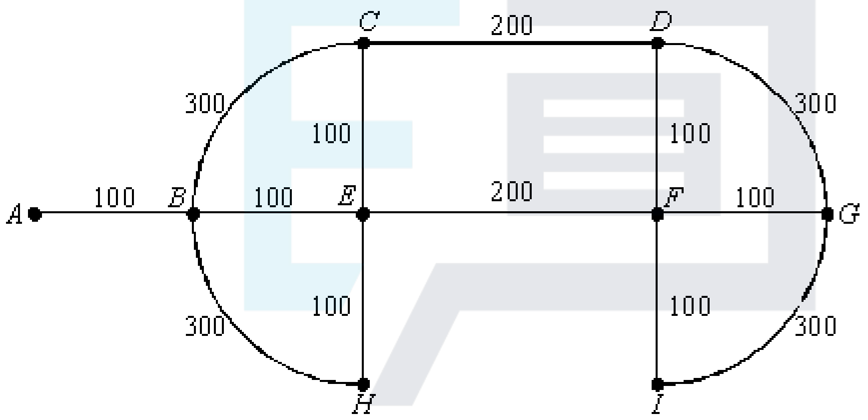
(b) Person **A** joins a telephone discount scheme which halves the cost of all phone calls to and from him. Without repeating the whole algorithm, state how this would affect your answer to part (a) and calculate the new minimum cost of notifying the whole family.

**(4)**
**(Total 9 marks)**

## Q19.

The edges of the network below represent roads in a small housing estate. There is only one road into and out of the estate, represented by edge $AB$. The lengths of the roads are shown in metres.

The total length of roads is 2300 m.




(a) A papergirl delivers to about 20% of the houses and therefore finds it worthwhile to cross from side to side of a road whilst making deliveries. Thus, she would **prefer** to walk along each road only once.



Use an appropriate algorithm to find the minimum total length of road along which she must walk if she starts and finishes her round at $A$. Indicate how you applied the algorithm.

**(4)**

(b) A postman delivers to about 80% of the houses. He finds it better to deliver to one side of a road at a time. He therefore needs to walk along each road at least twice. If he starts and finishes at $A$, find the minimum distance that he must walk, justifying your answer.

**(3)**

(c) Suppose that the postman acquires a bicycle. This means that he would still like to travel along each road twice, but in opposite directions, so that he is always riding on the correct side of the road. Will this mean that he must travel further? Justify your answer.

**(2)**
**(Total 9 marks)**